

## Workshop 10

### Problem 1

Keywords: Applications of the Mean Value Theorem

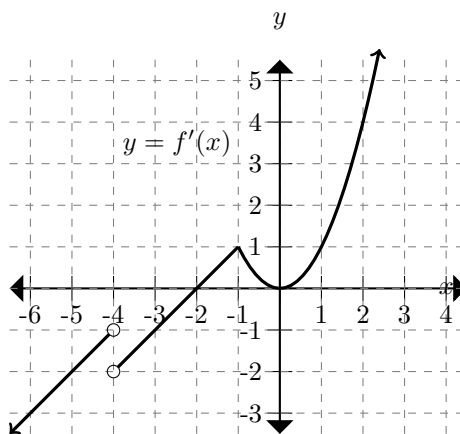
Your friend leaves your house at 9:00am and drives back to their apartment located 80 miles away. They call you from their home phone at 10:15am to confirm that they have arrived. They also tell you that they never drove over the speed limit, which is 60mph.

- (a) How can you tell that your friend is lying? Use a mathematical theorem to prove that at some point of their drive, your friend was going 64mph.
- (b) If your friend was telling the truth about not driving over the speed limit, what is the maximal distance from your house they could have reached at 10:15am?
- (c) Let  $f$  be a function continuous on  $[9, 10.25]$  and differentiable on  $(9, 10.25)$ . Assume that  $f(9) = 0$  and that  $f'(x) \leq 60$  for  $x$  in  $(9, 10.25)$ . According to the Mean Value Theorem, what is the largest value that  $f(10.25)$  can be? How does this question relate to the previous one? Explain in 1-2 English sentences.

### Problem 2

Keywords: Critical Points, Extreme Values, Intervals of Increase and Decrease, Graph Sketching

Suppose  $f$  is a continuous function on  $(-\infty, \infty)$  whose first derivative is sketched below. Note that  $f'$  is not defined at  $-4$ . Read carefully again: this is the graph of the **DERIVATIVE** of  $f$ .



- (a) Find the critical points for  $f$ .
- (b) Find the interval(s) where  $f$  is increasing and the interval(s) where  $f$  is decreasing.
- (c) Does  $f$  have any local extrema? If so, give the location of each local extremum and determine the nature of the local extremum.

- (d) Does  $f$  have any absolute extrema? If so, give the location of each absolute extremum and the nature of the absolute extremum. If this can't be determined, explain your reasoning.
- (e) Find the intervals where  $f$  is concave up, concave down and the inflection points of  $f$ .
- (f) We know that  $f$  is not differentiable at  $x = -4$ . What does the graph of  $f$  look like at that point? Choose between cusp, corner, vertical tangent, discontinuity or infinite oscillation.
- (g) Draw a possible sketch of the graph of  $y = f(x)$ . The scale of your sketch need not be accurate, only the general shape matters.

### Problem 3

Keywords: L'Hopital's Rule

Compute the following limits. If you use L'Hopital's rule, make sure to indicate where and say which indeterminate form allows you to use it.

(a)  $\lim_{x \rightarrow 0} \frac{\sin(6x) - 6x + 36x^3}{x^5}.$

(b)  $\lim_{x \rightarrow \infty} \left( \frac{x+1}{x+2} \right)^x.$

### Optional Problem *(just for fun, no need to submit)*

Let  $f$  be a function differentiable on  $[0, +\infty)$ . Assume that  $f(0) = 2$ , and that

$$2x - 4 \leq f'(x) \leq x^2 + 4x + 7.$$

Find two functions  $g(x)$  and  $h(x)$  such that  $g(x) \leq f(x) \leq h(x)$ .

*(Hint: use the Mean Value Theorem on the interval  $[0, x]$ . Then, find the absolute maximum of  $c^2 + 4c + 7$  and the absolute minimum of  $2c - 4$  for  $c$  in the interval  $[0, x]$ .)*